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Domain Specific Text Summarization for Hindi Using Transformer-Based Models

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***Abstract*—The rapid increase of digital information in India has generated an increasing demand for effective techniques to synthesize domain-specific literature in natural languages. In this research, the challenges and advancements in domain-specific text synthesis and summarization for Indian languages are explored. It summarizes deep learning along-with machine learning methods and focuses especially on customizing techniques to handle the linguistic complexity. The research highlights issues related to the lack of data available with low-resource languages. Furthermore, the study investigates the latest developments done in transformer-based models for text synthesis, including GPT-3 and BERT. The performance of the proposed model is examined in terms of different parameters such as precision (70.59%), recall (58.81%), and F1 score (62.34%).Besides addressing these issues, it promotes digital communication and information sharing that is inclusive in India.**

Keywords: BERT, GPT-3, Deep learning, Indian language, Summarization

I. Introduction

The online communication in India as well as rapid increase in digital content in coming years has demanded the need for sophisticated natural language processing (NLP) techniques that are appropriate for the linguistic subtleties of Indian languages. As the internet is becoming more significant source of information, the requirement of efficient native-language text production along with summary technology is increasing. This research examines the domain-specific developments and difficulties associated with text generation and summarization in the setting of Indian languages, with a focus on Hindi and Marathi. Natural language processing (NLP) finds Indian languages challenging due to their enormous vocabulary, intricate syntactic structures, and distinctive linguistic features. Thus, understanding these differences is essential to developing efficient domain-specific text summarizing algorithms.

This research focuses on text summarization while examining recent advancements in deep learning and machine learning techniques. It measures how well these methods manage domain-specific knowledge and enhance summarization models' functionality. It also highlights the challenges presented by low-resource languages, providing insights into strategies for dealing with the data scarcity issues that are prevalent in the Indian language scene. The study emphasizes the evolving field of text production, going beyond a mere synopsis. It assesses how transformer-based models, such GPT-3 and BERT, produce text in Indian languages that is both logical and culturally relevant. Understanding how these models interact with the linguistic diversity of India is essential to developing effective text generating systems.It has become crucial to extract vast amounts of things into concise yet practical summaries in an era where the amount of digital information is continuously growing. Textual data handling and comprehension are revolutionized by the use of Natural Language Processing (NLP). Natural language processing (NLP) is frequently used for text summarizing, which tackles the problem of reducing long texts, articles, or sections into shorter forms while maintaining the important features and main topics.

From early abstracts that were carefully chosen by humans to contemporary algorithms powered by computational linguistics, the art of summary spans centuries. Yet, the rise in online content production and the necessity for instant information access have made automatic and effective text summarizing methods necessary. With the development of NLP approaches, machines are now able to grasp, evaluate, and produce summaries that resemble those of humans.

The primary contribution of this research is

- The introduction of the IndicBART model from ai4Bharat for the text summarization job on the Hindi ILSUM/ILSUM-1.0 dataset
- To employ unique and comprehensive Hindi language features that generate more accurate text summarization than generic features.

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- To calculate text summarization convergence more quickly and accurately utilizing transformer-based models like BERT and GPT-3.
- To assess summarization outcomes using Recall-Oriented Understudy for Gisting Evaluation (ROUGE) metrics.

This is how the remainder of the paper is structured. The literature examination conducted in the past to identify research gaps and create an effective automatic text summarizing system is covered in Section 2.The comprehensive approach is presented in Section 3, covering sentence rating, automatic text creation processing, feature extraction, text pre-processing, and the summary generation phase. A dataset, evaluation metrics, and outcomes are included in the experimental setup that is shown in Section 4. The study's conclusions and future research initiatives are covered in Section 5.

II. Literature Review

Several studies have made significant contributions to the advancement of text summarization for Indian languages.Here, we discuss some notable studies:

Urlana, Ashok et al. [1] work investigates the application of pre-trained sequence-to-sequence models for summarizing Indian languages and English utilizing pre-trained Sequence-to-Sequence Models for Indian Language Summarization,. With the use of refined models such as PEGASUS and BART, the authors were able to obtain encouraging findings, with ROUGE scores for the Rouge L reaching 47.57%. In the research done by Tripathy, S.A[2], the redundancy and phrase structure are two problems that authors have suggested solving in their strategy for enhancing abstractive summarization. Their model has potential for practical use with its excellent accuracy and low training loss. The model has a Rouge-L score of 38.65%. Sarwadnya, V.V et al. [3] presented a Marathi text summary system. Their suggested model shows strong F-scores and accuracy, indicating that graph-based techniques work well for Indian languages. Their Rouge score for the Text Rank algorithm is 77.99%. Vijay et al. [4] address the need for automated text summarization in Hindi, achieving high precision and recall rates using statistical and linguistic features. Jain et al. [5] presented a genetic algorithm-based method for Hindi text summarizing with encouraging ROGUE scores of 58%. Taunk, Dhaval et al. [6] explored the application of multilingual transformers to Indian language summarization using Multilingual Transformers to improve abstract Summarization of Indian Languages. The results obtained a 44.5% improvement in performance through data augmentation. Verma, P. et al. [7] examines existing text summarization methods and show how NLP tools affect text summarization for Indian languages emphasizing the difficulties and prospects for further study. Chouk, M. et al. [8] provided extractive methods that achieve good accuracy rates for summarizing Marathi texts and classifying them using the Marathi headlines dataset. Agarwal, Arjit et al. [9] used IndicBART model, which obtained a ROUGE-1 F-score of 54.4% on the testing dataset, represents a notable improvement in text summarization for the Hindi language.

III. Methodology/Implementation

This research involves the application of the IndicBART model [9] from ai4bharat for a text summarization task on the ILSUM/ILSUM-1.0 dataset in Hindi [6]. The process as depicted in figure 1 involves several steps, including dataset loading, model and tokenizer initialization, feature extraction, dataset preparation, data collation, training setup, and model training. Each of these steps is crucial in preparing the data, setting up the model, and executing the training process.

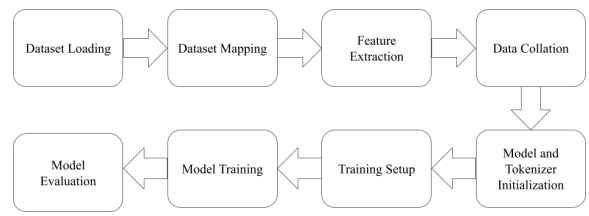

Fig. 1: Workflow of the Model

1. *Loading of Dataset*: The dataset namely ILSUM/ ILSUM-1.0, is utilized which is written in the Hindi language.It is loaded using the function load_dataset from the datasets library. This function allows loading of the dataset easily, which is the initial step in any machine learning or data analysis task.
2. *Initialization of Model*: In this study IndicBART model from ai4bharat is utilized. The model and its corresponding tokenism are loaded using the from pretrained method.This method loads a pre-trained model/tokenizer from either a local directory or a remote repository [6].

$$\text{Attention}(Q,K,V)= \text{softmax}\left(\frac{QK^T}{\sqrt{d_k}}\right)V \quad (1)$$

where

- Q is the query,

K and V stand for key along with value vectors, respectively,,

- d is the dimension of the key vectors.

3. *Feature Extraction*: To perform feature extraction,a function ('get_feature') is developed to tokenize the 'Article' and 'Summary' from the dataset as well as return the input IDs, attention masks, and labels. These are the features that the model will use to learn the mapping from articles to summaries.
4. *Dataset Preparation*: The get_feature function is applied to the dataset using the map method. This transforms the dataset by tokenizing the articles and summaries and extracting the necessary

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features. The format of the dataset is then set to 'torch' with columns 'input_ids', 'labels', and 'attention_mask'. This prepares the dataset for input into the model. Figure 2 and 3 displays the distribution of the number of words or characters present in both the articles and their corresponding summaries in the dataset

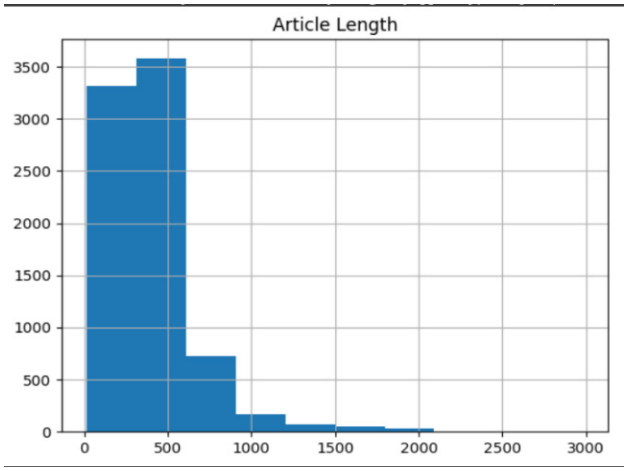

Fig. 2: Article lengths of the Dataset

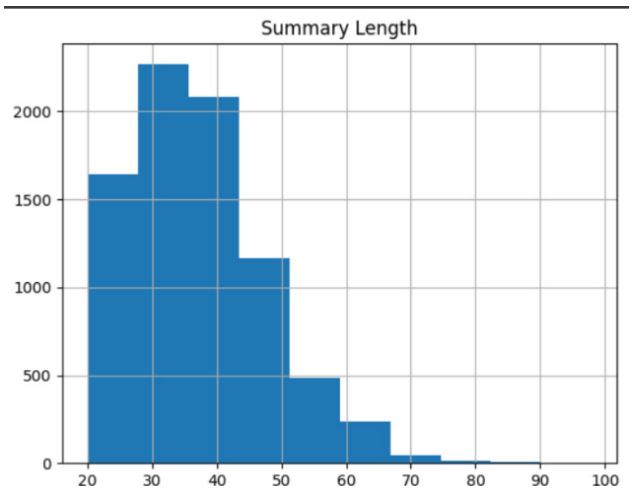

Fig. 3: Summary Length of the Dataset

5. *Data Collection*: A DataCollatorForSeq2Seq object is created with the tokenizer and model. This object is responsible for collating multiple data points into a batch. It ensures that the data fed into the model during training is in the correct format and structure.
6. *Training Setup*: The transformers library's Training Arguments class is used to define training arguments. The output directory, no. of training epochs, warmup steps, batch sizes, weight decay, logging steps, evaluation method, evaluation steps, save steps, and gradient accumulation stages are just a few of the training parameters that are specified by these options. These variables can be adjusted to maximize performance and regulate the training process. The equation (2) represents the gradient descent update rule commonly used in optimization for machine learning and deep learning models.

$$\theta = \theta - n^{\nabla\theta} \tag{2}$$

where
the parameters are denoted byθ,

- the learning rate byη,
- the gradient of the loss function $J(\theta)$ in relation to the parameters by $\nabla\theta J(\theta)$.

7. *Model Training*: The model, training arguments, tokenizer, data collator, and training and validation datasets are constructed as a Trainer object. The Trainer object abstracts away much of the complexity of training, making it easier to train models. The model is then trained using the train method of the Trainer object. This is the final step where the model learns to generate summaries from articles by minimizing the difference between its predictions and the actual summaries in the dataset. With an initial learning rate of 5e-5 and a batch size of 4 samples per device, the model was trained for a total of 4 epochs. The minimum requirements of these hyper parameters were contained in the training arguments (training_args).

IV. Results and Discussion

The outcomes of developing and testing the domain-specific text summarizing model for news stories in Hindi are mentioned in this section. The performance of the model will be examined in relation to the selected assessment criteria(ROUGE scores) and assess how well the data augmentation strategies were put into practice.

The model is evaluated using the ROUGE (Recall-Oriented Understudy for Gisting Evaluation) criteria, which are routinely used to measure automatic summarization and machine translation systems. The ROUGE metrics compare an automatically generated summary (the hypothesis) to a reference summary (the original text).

Table I: Rouge Score Evaluation

	R1	R2	RL
Recall	0.581395	0.428571	0.588139
Precision	0.735294	0.538461	0.705882
F1-Score	0.649350	0.477272	0.623376

The proposed model achieved the following ROUGE scores as given below:

ROUGE-1: The hypothesis and reference's unigrams, or single words, are compared using this parameters. It is observed that model performs well with an F1 score of 64.93%, recall of 58.13%, and accuracy of 73.52%.

ROUGE-2: In the hypothesis and reference, bigrams (two consecutive words) are compared using this measure. With an F1 score of 47.73%, recall of 42.86%, and accuracy of 53.85%, our model performed well.

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ROUGE-L: Based on longest common sub-sequence (LCS) data, this measure compares the longest matching word sequence. With an F1 score of 62.34%, recall of 58.81%, and accuracy of 70.59%, our model performed well.

These findings in figure 4 show that, while there is potential for improvement, our model does rather well in producing summaries that are comparable to the source text, especially when it comes to matching bi-grams and longer word sequences.

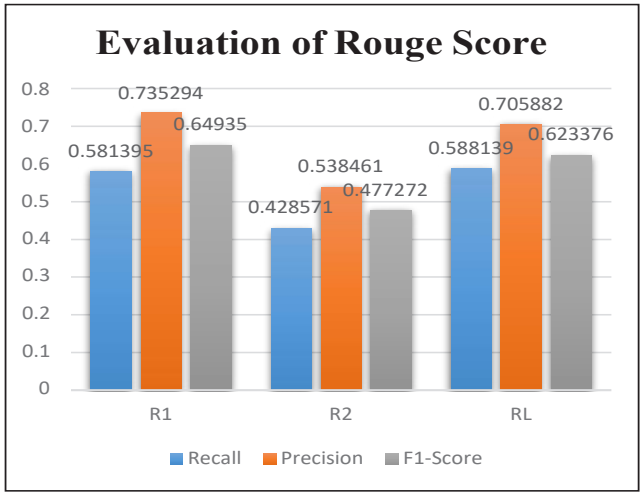

Fig. 4: Evaluation of Rouge Score

According to our research, the IndicBART model from the work of Agarwal, Arjit et al. [9] is not as good as our model. With R1, R2, and RL F1 scores of 64.93%, 47.73%, and 62.34%, respectively, our model produced excellent results. In comparison, the IndicBART model's scores were 54.43% for ROUGE-1 and 44.33% for ROUGE-2, with the ROUGE-L score not explicitly provided. These results highlight that our model outperformed the IndicBART model by achieving higher ROUGE scores across all metrics, demonstrating enhanced summarization capabilities for Hindi text.

V. Conclusion

The study examines challenges and advancements in domain- specific text summarization for an Indian language, emphasizing the need for tailored techniques. It highlights the use of models like TextRank and IndicBART, and the encouraging results of the IndicBART model for Hindi text summarization based on ROUGE metrics. The paper also explores the potential of transformer-based models like GPT-3 and BERT. Future research will focus on advanced transformer models, attention mechanisms, hierarchical and multimodal summarization, human evaluation studies, task-oriented evaluation metrics, bias detection, attention visualization, contrastive explanation methods, personalization of summaries, real-time summarization applications, and potential adaptation for multilingual summarization and other low-resource languages. The goal is to advance text summarization and promote inclusive digital communication. This research advances the field of text summarization for Hindi, potentially aiding in the development of inclusive digital communication tools in multilingual contexts.

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